

Registration of 3D Ultrasound Images using Iterative Closest Point

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Abstract. This study implements rigid registration of 3D ultrasound images using Iterative Closest Point (ICP) algorithms using Sobel filter for edge detection feature extraction. Random transformation is applied to simulate real world applications, with evaluation metrics such as Euclidean distance, Root Mean Square Error (RMSE), and Dice Similarity Coefficient (DSC) for alignment accuracy. Results show moderate feature overlap and accuracy in tandem with ICP sensitivity to transformation parameters and preprocessing. The study underscores the need for a more effective registration technique and feature extraction method

Keywords: Image Registration, Iterative Closest Point, Feature Extraction.

1 Introduction

The analysis of medical images is crucial in studying diseases and illnesses. Understanding the correspondence between spatial locations in medical images allows for direct comparisons between anatomical structures [1]. This information transfer allows for the identification of various anatomical boundaries in different images [1][2]. Image registration is the process of computing and aligning these different correspondences. Accurate image registration enhances visualization, diagnostic accuracy and surgical precision [3]. The Iterative Closest Point (ICP) algorithm is the most known method of image registration due to its simplicity and efficiency in aligning through point clouds [4]. The ICP algorithm iterates and refines a transformation to close the distance between two sets of correlating points [5].

3D ultrasound images are widely used in medical imaging due to their non-invasive and real-time visual feedback [6], but noise and artifacts are impeding registration accuracy. Enhanced image features like edges and gradients can reduce computational complexity and focus on more important structures [7][8]. This project aims to design a feature extraction method in tandem with ICP image registration for feature detection in a 3D ultrasound image. This study aims to improve alignment accuracy, test parameters and assess registration performance qualitatively and quantitatively and discuss on which approach is best for the most viable results.

2 Methods

The SimpleITK library is installed in the “mphy0030-cw” environment. It loads and extracts metadata from the GIPL document [9], including voxel spacing, origin, spacing and direction, for spatial interpretation and analysis [8]. The image features are extracted, including edge feature extraction used in Sobel filtering to identify and differentiate between different voxel intensities [10]. This technique is useful for point and feature based registration in ICP [11]. Sobel filtering weighs central pixels more heavily making it more resilient to noise (speckle noise) which ultrasound images are known for having [12].

A function is developed to simulate real-life misalignments in medical images by randomizing transformations (translation and rotation), a method highly used in algorithm testing [4]. Rigid transformation of the image is done through resampling, preserving its original shape and size [13]. Resampling the image warps and transforms it with specified translation and rotation [14]. A nearest neighbour interpolation is used to prevent artifacts from surfacing [15] whilst also preserving intensity values followed by Singular Value Decomposition (SVD). SVD is a robust method for handling rotation of medical images, minimizing alignment error especially for noisy and imperfect data [16].

ICP is implemented to minimize the sum of squared distance between corresponding points ensuring convergence to a local minimum representing the best alignment [7]. ICP starts with a transformation of the moving set, then each point in the moving set finds the nearest point of the fixed set.

3 Experiments

The ICP accuracy is evaluated through a try-and-error process using random translation vectors and rotation angles of Euler transformations [4]. Testing the transformation parameters helps to assess the algorithm’s performance. A translation range of 10-30mm is sufficient for representing realistic displacements between scans [17]. Larger translations can shift the edge features out of the image frame. A rotation range between 15-45 degrees is a realistic mimicry of angular deviations of ultrasound probe movements [3]. Angles above 45 degrees can increase computational burden. Smaller transformation ranges are easier to compute but less viable for pre-clinical testing.

SimpleITK’s Euler3DTransform is used for rigid transformation, simulating the moving image and registering it back to the original image. It can modify spatial coordinates of image voxels without changing pixel intensity values [18]. This step is crucial in clinical settings where images are scanned from various positions. Edge points are extracted from both images to create feature sets for registration [8], reducing data complexity by isolating important edge points [19].

ICP aligns the transformed image’s edge points to the original to find the best fit with iterations ranging from 20-100 iterations for general point cloud alignment tasks [4]. The registration quality is assessed based on the results of the following result metric

- **Final Translation:** 3D shifts (x, y, z) in mm to align moving to fixed image.
Importance: Indicates corrected spatial displacement.
- **Final Rotation Matrix:** 3D rotation applied to align images.
Importance: Reflects angular correction magnitude.
- **Euclidean Distance [20]:** Distance between corresponding points in fixed and moving images.
Importance: Lower values indicate better alignment.
- **Root Mean Square Error (RMSE) [21]:** Average squared Euclidean distances between corresponding points.
Importance: Quantifies overall registration accuracy.
- **Dice Similarity Coefficient (DSC) [22]:** Overlap measure between fixed and registered images (0 to 1).
Importance: Higher values show better segmentation and alignment.

4 Results

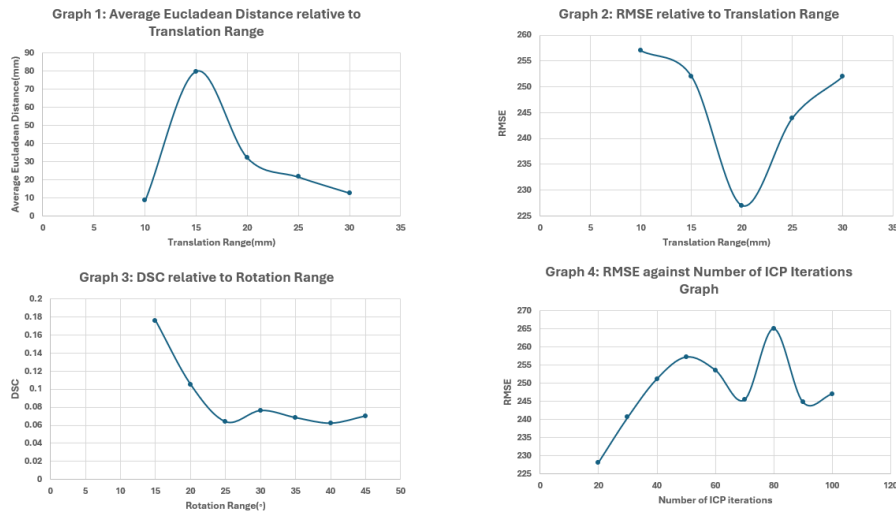


Fig. 1. Graphs of Result Metrics

5 Discussion

Graph 1 shows how the translation magnitude significantly impacts point matching accuracy. The high spike around 15mm translation range shows how initial misalignment greatly affects ICP's initial performance but more corrections are made below and above 15mm showing better alignments with reduced Euclidean distance. This proves that ICP is sensitive to initial misalignments, depending on good initial estimates for effective registration [16]. In Graph 2, the U-shaped curve signifies that the algorithm

works best in the 20mm range and not any other. ICP may face struggles and issues with other translation ranges without pre-registration [4], emphasizing the need for careful preprocessing for larger misalignments. DSC values in Graph 3 highlight that small rotations are optimal for convergence and overlap between fixed and moving images [23]. The graph suggests that rigid transformations for larger rotation angles require a more robust alignment technique and preprocessing. The final graph shows a non-linear pattern on local minima trap encountered by the algorithm [24], and mitigation strategies like outlier rejection can be applied [25]. A definite number of iterations cannot be concluded from the graph where future boundary criteria can be established for a more appropriate setting.

6 Conclusion

This coursework has explored the use of feature based rigid registration using ICP to align the 3D ultrasound image. Real world clinical scenarios are simulated to evaluate its effectiveness with random transformations and edge feature detection and extraction was done using Sobel filter. Result metrics such as Euclidean distance, RMSE and DSC all are important key data providing insights of the registration performance. Although DSC shows very little overlap, RMSE values show obvious alignment accuracy and outliers are introduced with variability. This just shows that parameter selection of translation and rotation should be done carefully as they impact image accuracy and convergence. Future work could focus on enhancing robustness with more advanced feature extraction and data structure.

7 References

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